CBSE Class 10 Mathematics — Statistics

Time Allowed: 3 Hours **Maximum Marks: 80**

General Instructions:

1. This question paper contains $_{questions. All questions are compulsory.}$

1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. The mean of the first 5 natural numbers is:

(A) (A) 2
(B) (B) 3
(C) (C) 4
(D) (D) 5

2. Find the mode of the following data: 2, 3, 4, 3, 5, 3, 6, 3.

(A) (A) 2
(B) (B) 4
(C) (C) 3
(D) (D) 6

3. If the mean of $x, x + 2, x + 4, x + 6, x + 8$ is 24, find the value of x.

(A) (A) 20
(B) (B) 22
(C) (C) 24
(D) (D) 26

4. The median of first 7 prime numbers is:

(A) (A) 5
(B) (B) 7
(C) (C) 11
(D) (D) 13

5. For a given distribution, if Mean = 20 and Mode = 20, find the Median using the empirical relationship.

(A) (A) 15

(B) (B) 18
(C) (C) 20
(D) (D) 25

6. Assertion (A): The mean of the first five natural numbers is 3. Reason (R): The mean of a set of observations is the sum of observations divided by the total number of observations.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true

7. Assertion (A): The empirical relationship between mean, median, and mode is 3Median = Mode + 2Mean. Reason (R): For a symmetric distribution, mean, median, and mode are equal.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true

8. Assertion (A): The mode of the data 2, 3, 4, 3, 2, 3 is 3. Reason (R): Mode is the observation that occurs most frequently.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true

9. $Assertion (A) : If the mean of a distribution is 20 and the sum of observations is 100, then the number of$
$The sum of observations = Mean \times Number of observations.$

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): The median of the data 5, 8, 12, 16, 20 is 12. Reason (R): The median is always the middle value when the data is arranged in ascending order.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. The mean of 5 observations is 15. If each observation is increased by 3, find the new mean.

2. Find the class mark of the class interval 150-200.

3. If the mode of a data set is 45 and the mean is 27, find the median using the empirical relationship.

4. The frequency of a class is 12 and the total number of observations is 40. Find the relative frequency of this class.

5. Given the data: 12, 15, 18, 12, 15, 12, 18. Find the range of the data.

Section D: Long Answer (4-5 marks each)

1. The mean of the following frequency distribution is 50. Find the value of p if the sum of frequencies is 100: — Class — 0-20 — 20-40 — 40-60 — 60-80 — 80-100 — ———————————— — Frequency — 17 — p — 32 — p — 19 —

2. The following distribution shows the daily pocket allowance of children of a locality. If the mean pocket allowance is Rs. 18, find the missing frequency f: — Daily allowance (in Rs) — 11-13 — 13-15 — 15-17 — 17-19 — 19-21 — 21-23 — 23-25 — ———————————— — Number of children — 7 — 6 — 9 — 13 — f — 5 — 4 —

3. Find the median of the following data: — Marks — 0-10 — 10-20 — 20-30 — 30-40 — 40-50 — ———————————— — Number of students — 5 — 15 — 20 — 7 — 3 —

4. The following table shows the ages of 100 patients admitted in a hospital during a year. Find the mode of the ages: — Age (in years) — 5-15 — 15-25 — 25-35 — 35-45 — 45-55 — 55-65 — ———————————— — Number of patients — 6 — 11 — 21 — 23 — 14 — 5 —

5. A survey conducted on 20 households in a locality by a group of students resulted in the following frequency table for the number of family members in a household: — Family size — 1-3 — 3-5 — 5-7 — 7-9 — 9-11 — ———————————— — Frequency — 7 — 8 — 2 — 2 — 1 — Find the mode of this data.

6. The mean of the following frequency distribution is 50. Find the values of p and q, if the sum of all frequencies is 120.

— Class Interval — 0-20 — 20-40 — 40-60 — 60-80 — 80-100 — — :— — :— — :— — :— — :— — :— — — Frequency — 17 — p — 32 — q — 19 —

7. The median of the following data is 525. Find the values of x and y, if the total frequency is 100.

— Class Interval — 0-100 — 100-200 — 200-300 — 300-400 — 400-500 — 500-600 — 600-700 — 700-800 — 800-900 — 900-1000 — — :— — :— — :— — :— — :— — :— — :— — :— — :— — :— — :— — — Frequency — 2 — 5 — x — 12 — 17 — 20 — y — 9 — 7 — 4 —

8. The following distribution gives the daily income of $50 workers of a factory. Convert the distributional$
$120|120-140|140-160|160-180|180-200|| : ---| : ---| : ---| : ---| : ---| : ---||Number of workers|12|14|8|6|10|$

9. Find the mode of the following frequency distribution. If the mean of the same data is 28.6, verify the empirical relationship between mean, median, and mode.

— Class — 0-10 — 10-20 — 20-30 — 30-40 — 40-50 — — :— — :— — :— — :— — :— — :— — — Frequency — 5 — 8 — 20 — 15 — 7 —